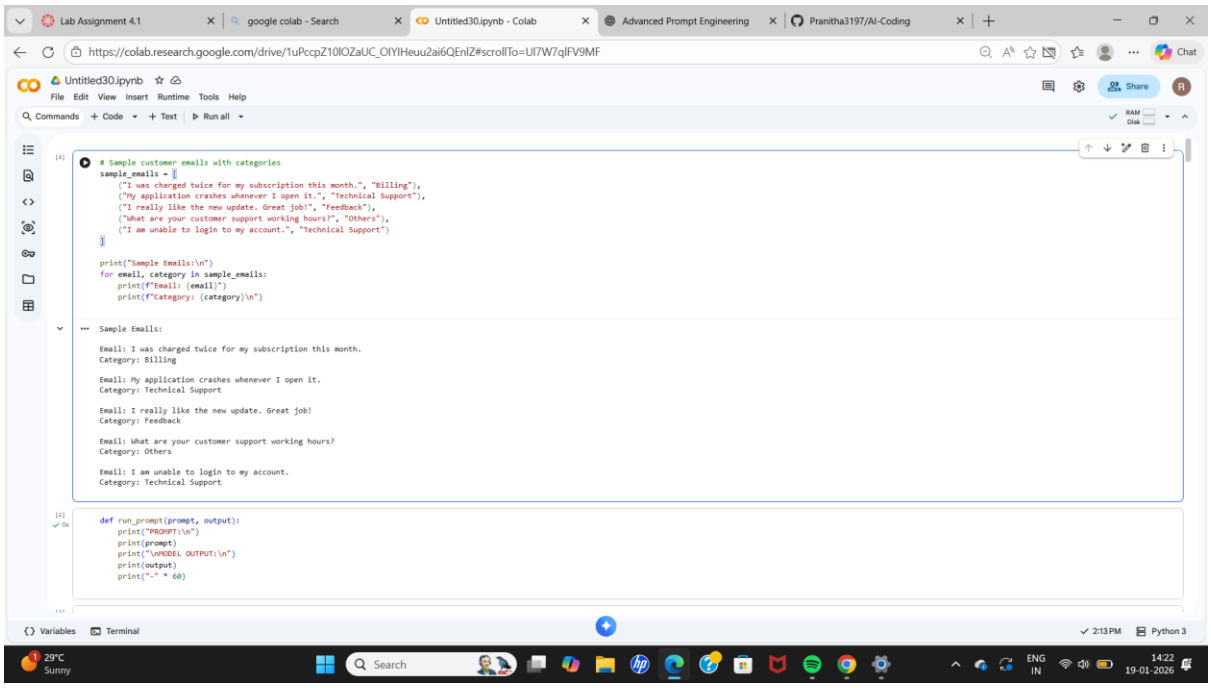
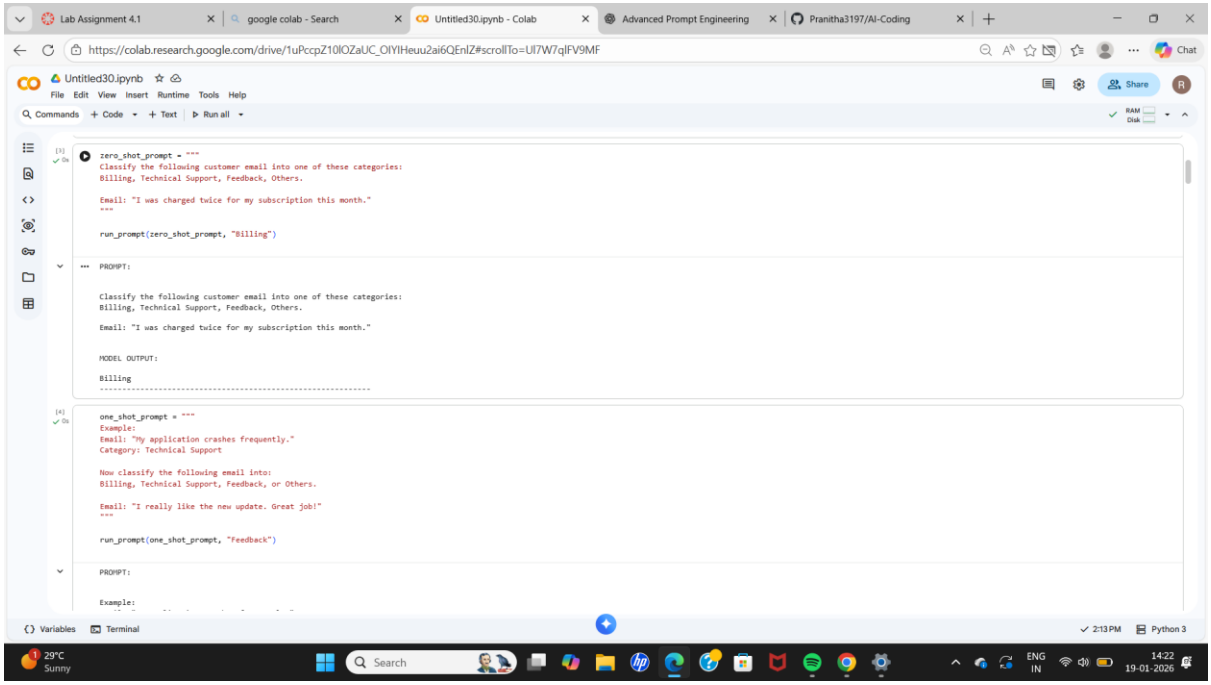
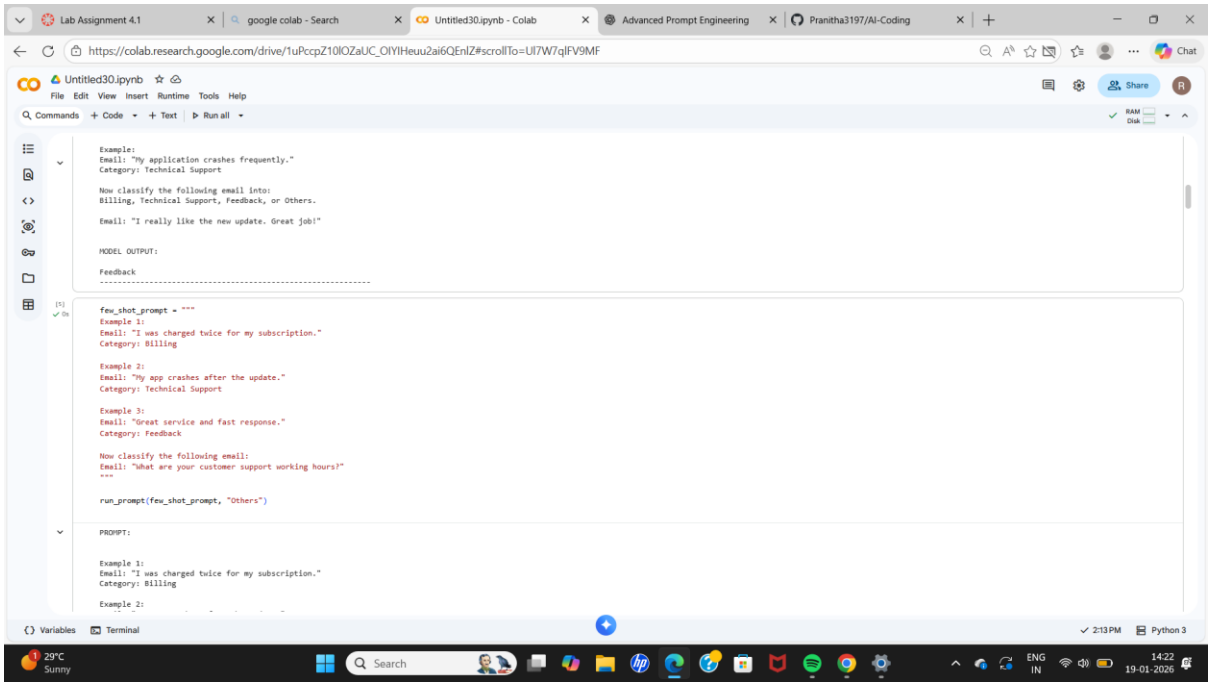
LAB ASSIGNMENT 4.1

NAME : R.PRANITHA REDDY 2303A52248 SUBJECT : AI ASSISTED CODING



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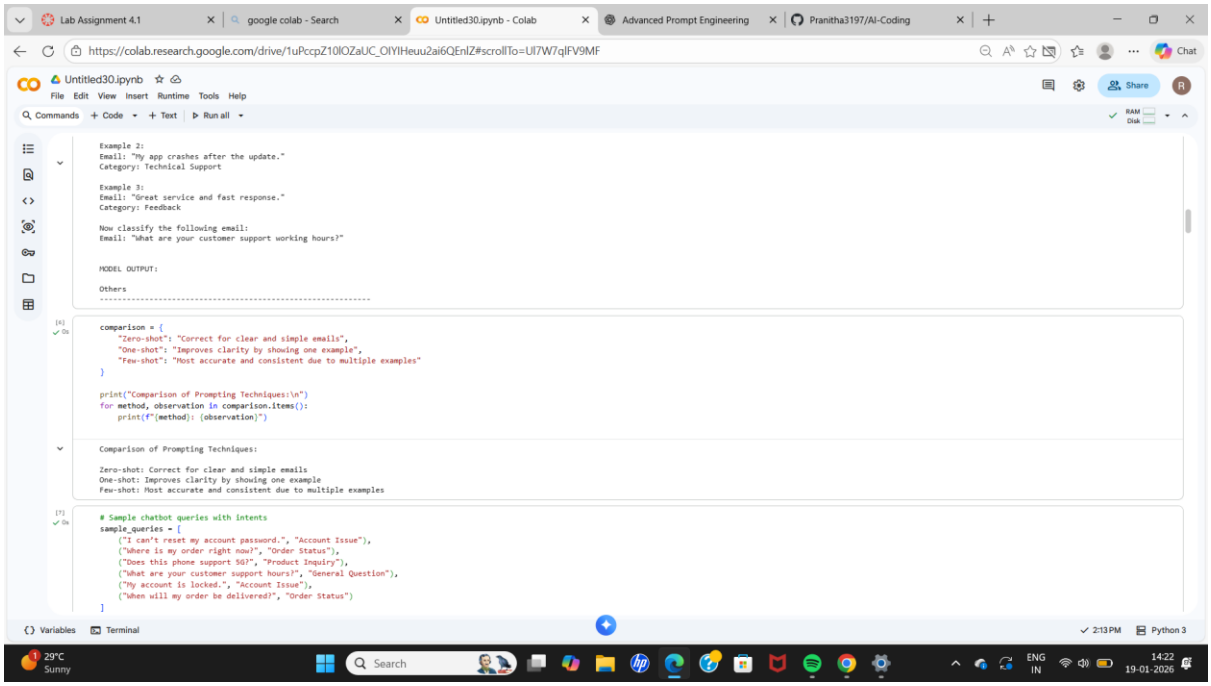
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Example:
Email: "My application crashes frequently."
Category: Technical Support
Now classify the following email into:
Billing, Technical Support, Feedback, or Others.
Email: "I really like the new update. Great job!"
MODEL OUTPUT:
Feedback
few_shot_prompt = """
Example 1:
Email: "I was charged twice for my subscription."
Category: Billing
Example 2:
Email: "My app crashes after the update."
Category: Technical Support
Example 3:
Email: "Great service and fast response."
Category: Feedback
Now classify the following email:
Email: "What are your customer support working hours?"
"""
run_prompt(few_shot_prompt, "Others")
PROMPT:
Example 1:
Email: "I was charged twice for my subscription."
Category: Billing
Example 2:
Variables
Terminal
2:13 PM
Python 3
29°C
Sunny
Search
ENG
IN
14:22
19-01-2026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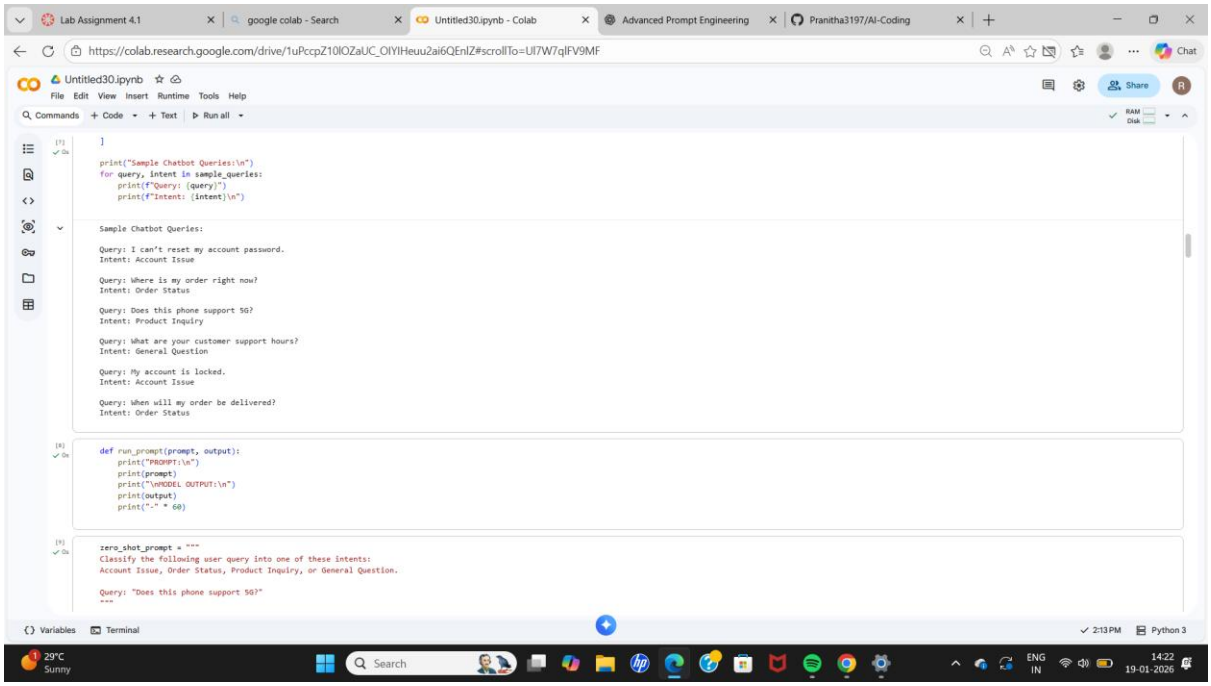
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Example 2:
Email: "My app crashes after the update."
Category: Technical Support
Example 3:
Email: "Great service and fast response."
Category: Feedback
Now classify the following email:
Email: "What are your customer support working hours?"
MODEL OUTPUT:
Others
comparison = {
"Zero-shot": "Correct for clear and simple emails",
"One-shot": "Improves clarity by showing one example",
"Few-shot": "Most accurate and consistent due to multiple examples"
}
print("Comparison of Prompting Techniques:\n")
for method, observation in comparison.items():
print(f"{method}: {observation}")
Comparison of Prompting Techniques:
Zero-shot: Correct for clear and simple emails
One-shot: Improves clarity by showing one example
Few-shot: Most accurate and consistent due to multiple examples
# Sample chatbot queries with intents
sample_queries = [
("I can't reset my account password.", "Account Issue"),
("Where is my order right now?", "Order Status"),
("Does this phone support 5G?", "Product Inquiry"),
("What are your customer support hours?", "General Question"),
("My account is locked.", "Account Issue"),
("When will my order be delivered?", "Order Status")
]
Variables
Terminal
2:13 PM
Python 3
29°C
Sunny
Search
ENG
IN
14:22
19-01-2026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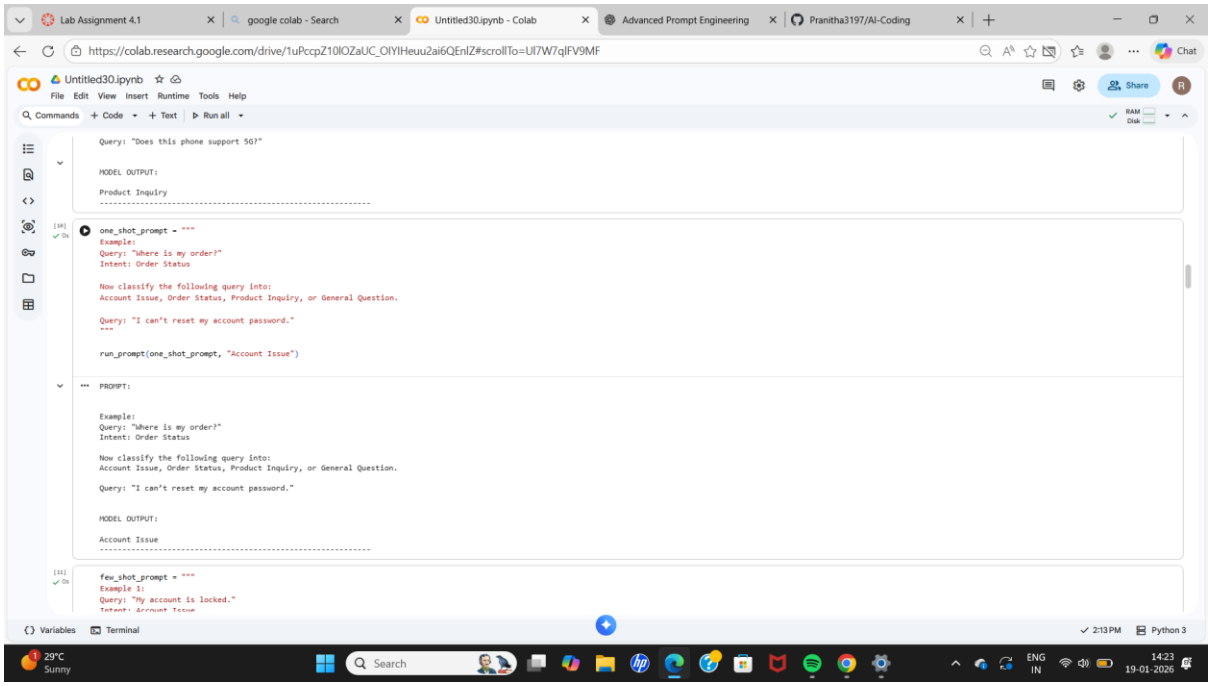
Untitled30.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
]
print("Sample Chatbot Queries:\n")
for query, intent in sample_queries:
print(f"Query: {query}")
print(f"Intent: {intent}\n")
Sample Chatbot Queries:
Query: I can't reset my account password.
Intent: Account Issue
Query: Where is my order right now?
Intent: Order Status
Query: Does this phone support 5G?
Intent: Product Inquiry
Query: What are your customer support hours?
Intent: General Question
Query: My account is locked.
Intent: Account Issue
Query: When will my order be delivered?
Intent: Order Status
def run_prompt(prompt, output):
print("PROMPT:\n")
print(prompt)
print("\nMODEL OUTPUT:\n")
print(output)
print("-" * 60)
zero_shot_prompt = """
Classify the following user query into one of these intents:
Account Issue, Order Status, Product Inquiry, or General Question.
Query: "Does this phone support 5G?"
"""
Variables Terminal
2:13 PM Python 3
29°C Sunny
Search
ENG IN
14:22 19-01-2026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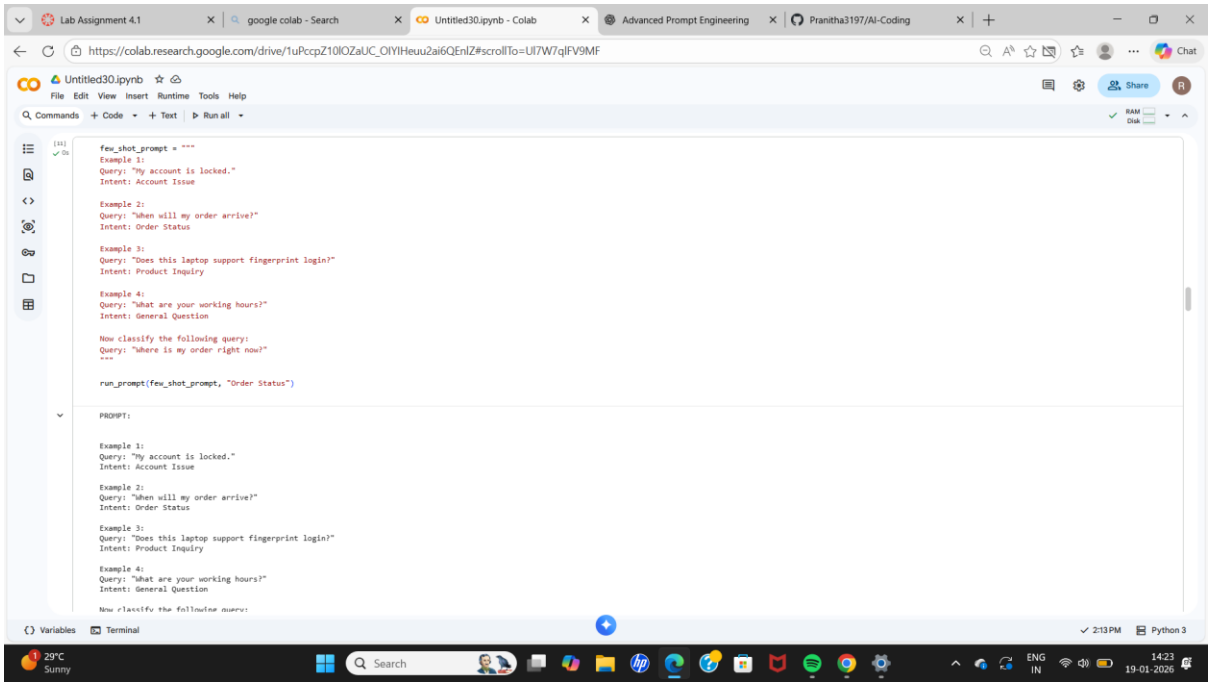
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Query: "Does this phone support 5G?"
MODEL OUTPUT:
Product Inquiry
one_shot_prompt = """
Example:
Query: "Where is my order?"
Intent: Order Status
Now classify the following query into:
Account Issue, Order Status, Product Inquiry, or General Question.
Query: "I can't reset my account password."
"""
run_prompt(one_shot_prompt, "Account Issue")
PROMPT:
Example:
Query: "Where is my order?"
Intent: Order Status
Now classify the following query into:
Account Issue, Order Status, Product Inquiry, or General Question.
Query: "I can't reset my account password."
MODEL OUTPUT:
Account Issue
few_shot_prompt = """
Example 1:
Query: "My account is locked."
Variables Terminal
2:13 PM Python 3
29°C Sunny
Search
ENG IN
14:23 19-01-2026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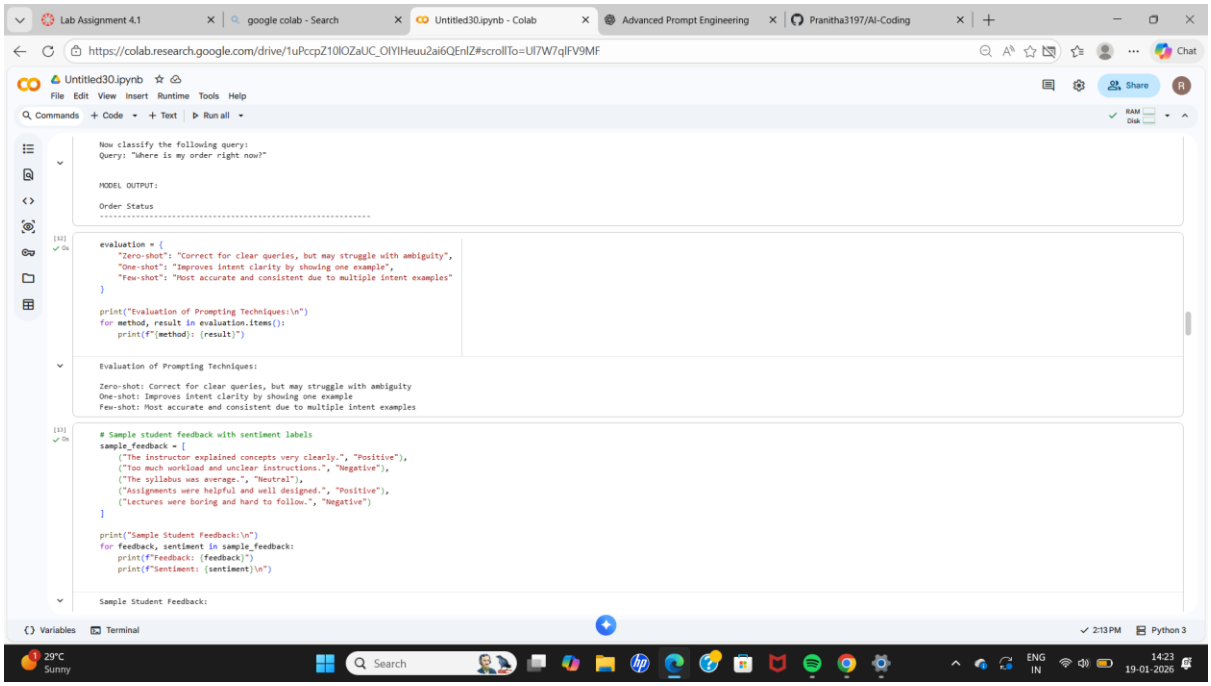
Untitled30.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
few_shot_prompt = """
Example 1:
Query: "My account is locked."
Intent: Account Issue
Example 2:
Query: "When will my order arrive?"
Intent: Order Status
Example 3:
Query: "Does this laptop support fingerprint login?"
Intent: Product Inquiry
Example 4:
Query: "What are your working hours?"
Intent: General Question
Now classify the following query:
Query: "Where is my order right now?"
"""
run_prompt(few_shot_prompt, "Order Status")
PROMPT:
Example 1:
Query: "My account is locked."
Intent: Account Issue
Example 2:
Query: "When will my order arrive?"
Intent: Order Status
Example 3:
Query: "Does this laptop support fingerprint login?"
Intent: Product Inquiry
Example 4:
Query: "What are your working hours?"
Intent: General Question
Variables Terminal
2:13 PM Python 3
29°C Sunny
Search
ENG IN
14:23 19-01-2026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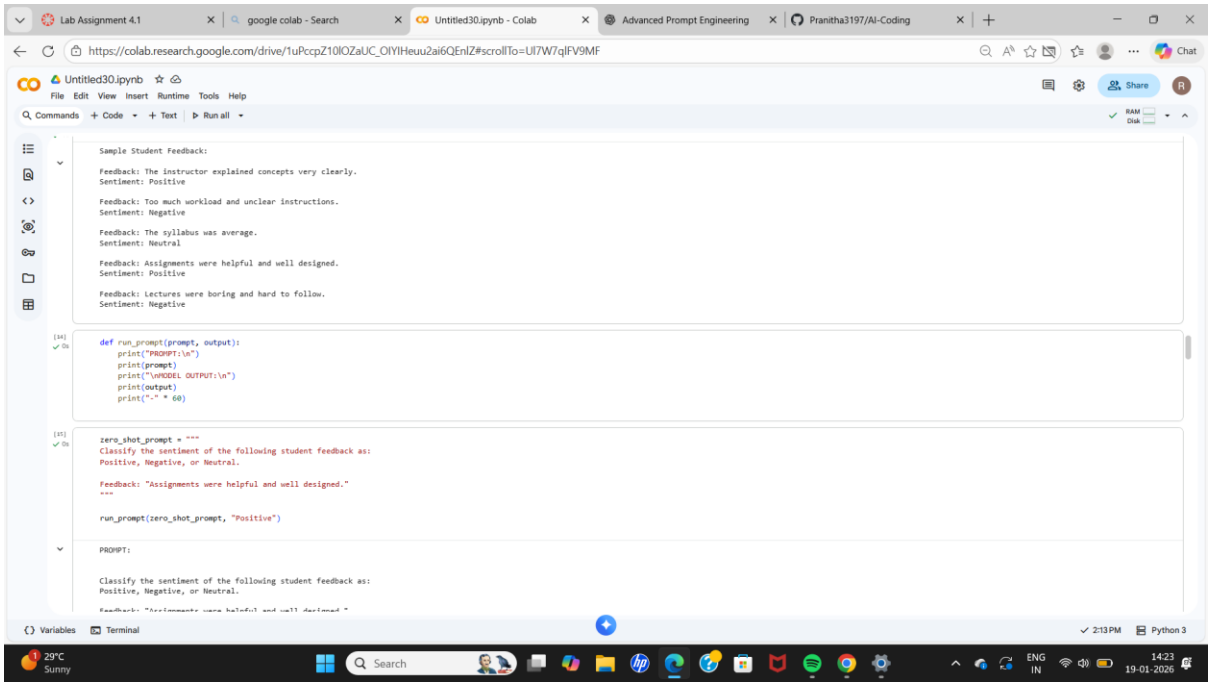
Commands + Code + Text Run all
Now classify the following query:
Query: "Where is my order right now?"
MODEL OUTPUT:
Order Status
evaluation = {
"Zero-shot": "Correct for clear queries, but may struggle with ambiguity",
"One-shot": "Improves intent clarity by showing one example",
"Few-shot": "Most accurate and consistent due to multiple intent examples"
}
print("Evaluation of Prompting Techniques:\n")
for method, result in evaluation.items():
print(f"{method}: {result}")
Evaluation of Prompting Techniques:
Zero-shot: Correct for clear queries, but may struggle with ambiguity
One-shot: Improves intent clarity by showing one example
Few-shot: Most accurate and consistent due to multiple intent examples
# Sample student feedback with sentiment labels
sample_feedback = [
("The instructor explained concepts very clearly.", "Positive"),
("Too much workload and unclear instructions.", "Negative"),
("The syllabus was average.", "Neutral"),
("Assignments were helpful and well designed.", "Positive"),
("Lectures were boring and hard to follow.", "Negative")
]
print("Sample Student Feedback:\n")
for feedback, sentiment in sample_feedback:
print(f"Feedback: {feedback}")
print(f"Sentiment: {sentiment}\n")
Sample Student Feedback:
Variables Terminal
2:13 PM Python 3
29°C Sunny
Search
ENG IN
14:23 19-01-2026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Share
Sample Student Feedback:
Feedback: The instructor explained concepts very clearly.
Sentiment: Positive
Feedback: Too much workload and unclear instructions.
Sentiment: Negative
Feedback: The syllabus was average.
Sentiment: Neutral
Feedback: Assignments were helpful and well designed.
Sentiment: Positive
Feedback: Lectures were boring and hard to follow.
Sentiment: Negative
def run_prompt(prompt, output):
print("PROMPT:\n")
print(prompt)
print("\nMODEL OUTPUT:\n")
print(output)
print("-" * 60)
zero_shot_prompt = """
Classify the sentiment of the following student feedback as:
Positive, Negative, or Neutral.
Feedback: "Assignments were helpful and well designed."
"""
run_prompt(zero_shot_prompt, "Positive")
PROMPT:
Classify the sentiment of the following student feedback as:
Positive, Negative, or Neutral.
Variables
Terminal
2:13 PM
Python 3
29°C
Sunny
Search
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IN
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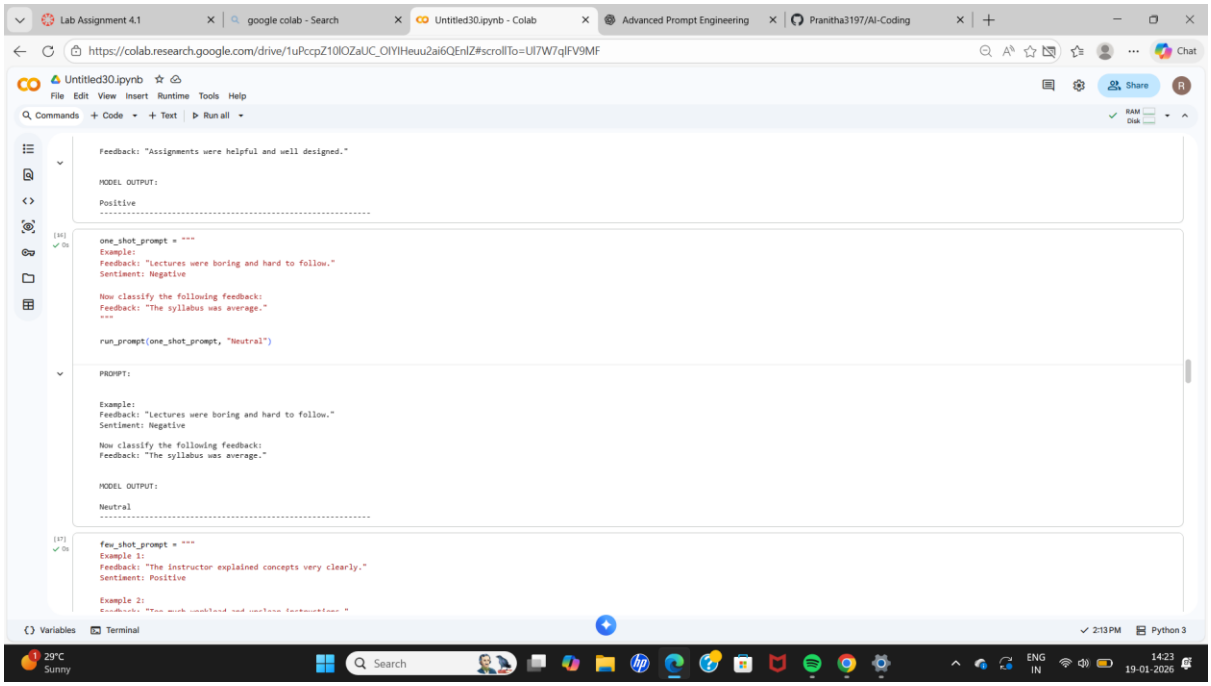
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File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Share
Feedback: "Assignments were helpful and well designed."
MODEL OUTPUT:
Positive
one_shot_prompt = """
Example:
Feedback: "Lectures were boring and hard to follow."
Sentiment: Negative
Now classify the following feedback:
Feedback: "The syllabus was average."
"""
run_prompt(one_shot_prompt, "Neutral")
PROMPT:
Example:
Feedback: "Lectures were boring and hard to follow."
Sentiment: Negative
Now classify the following feedback:
Feedback: "The syllabus was average."
MODEL OUTPUT:
Neutral
few_shot_prompt = """
Example 1:
Feedback: "The instructor explained concepts very clearly."
Sentiment: Positive
Example 2:
Variables
Terminal
2:13 PM
Python 3
29°C
Sunny
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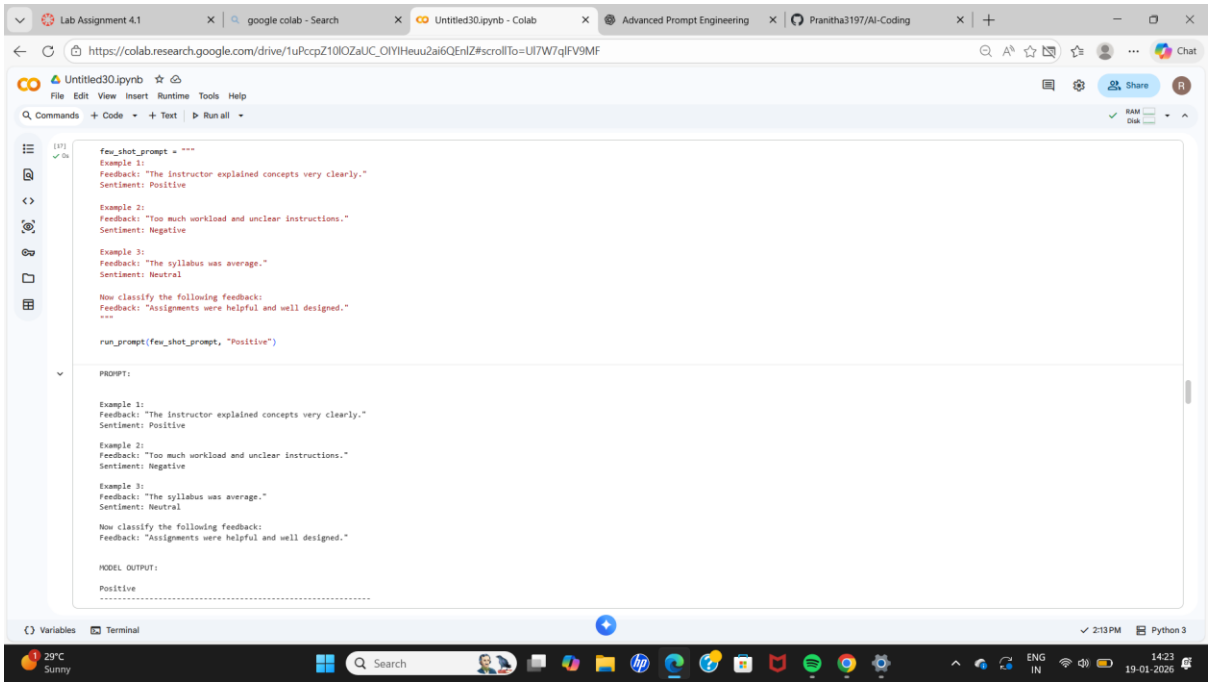
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Untitled30.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
few_shot_prompt = """
Example 1:
Feedback: "The instructor explained concepts very clearly."
Sentiment: Positive
Example 2:
Feedback: "Too much workload and unclear instructions."
Sentiment: Negative
Example 3:
Feedback: "The syllabus was average."
Sentiment: Neutral
Now classify the following feedback:
Feedback: "Assignments were helpful and well designed."
"""
run_prompt(few_shot_prompt, "Positive")
PROMPT:
Example 1:
Feedback: "The instructor explained concepts very clearly."
Sentiment: Positive
Example 2:
Feedback: "Too much workload and unclear instructions."
Sentiment: Negative
Example 3:
Feedback: "The syllabus was average."
Sentiment: Neutral
Now classify the following feedback:
Feedback: "Assignments were helpful and well designed."
MODEL OUTPUT:
Positive
Variables
Terminal
2:13 PM
Python 3
29°C
Sunny
Search
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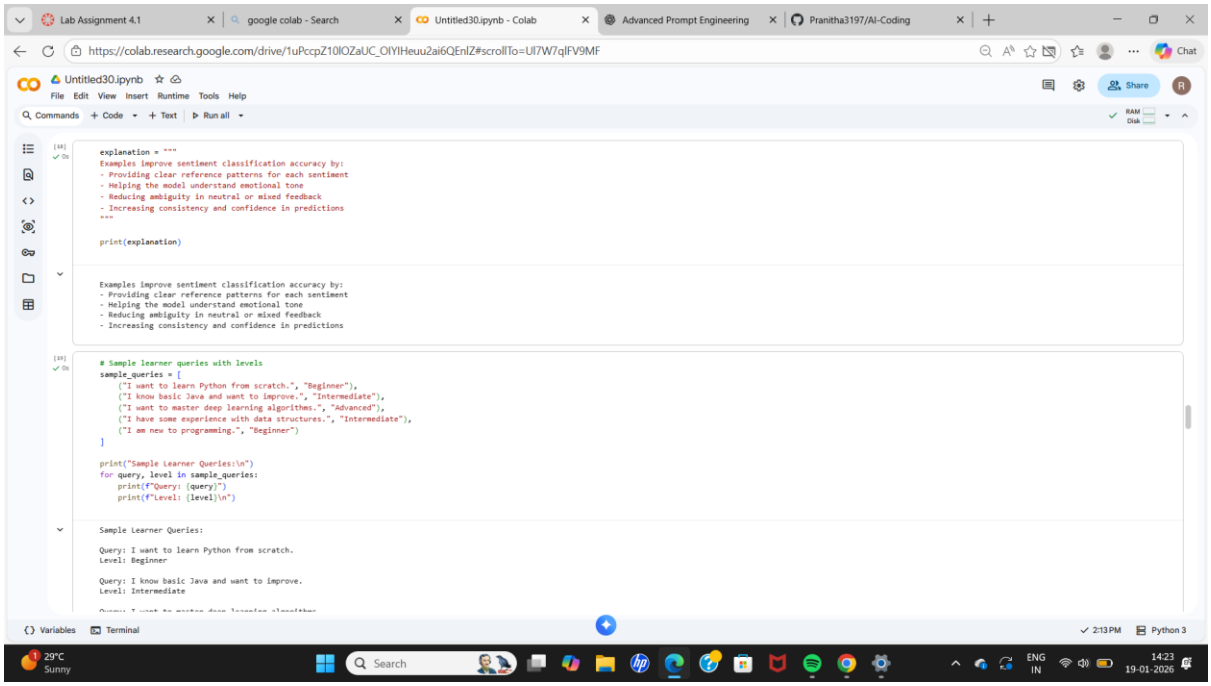
14:23
19-01-2026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Untitled30.ipynb
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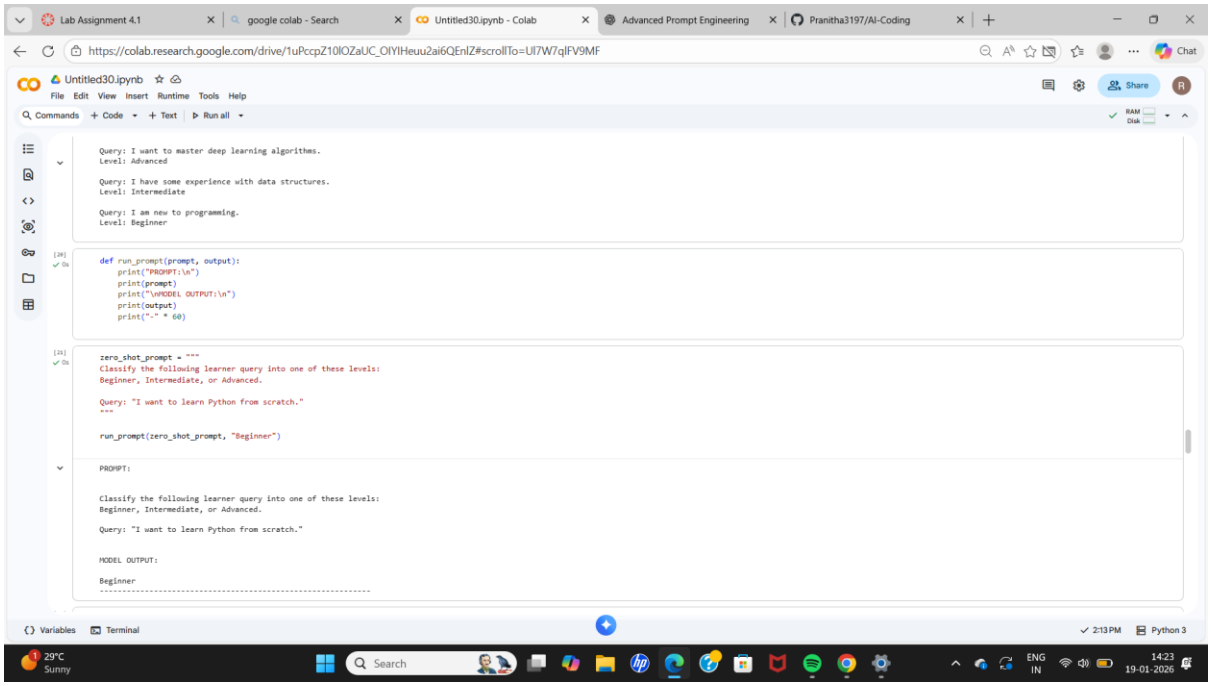
Commands + Code + Text Run all
explanation = """
Examples improve sentiment classification accuracy by:
- Providing clear reference patterns for each sentiment
- Helping the model understand emotional tone
- Reducing ambiguity in neutral or mixed feedback
- Increasing consistency and confidence in predictions
"""
print(explanation)
Examples improve sentiment classification accuracy by:
- Providing clear reference patterns for each sentiment
- Helping the model understand emotional tone
- Reducing ambiguity in neutral or mixed feedback
- Increasing consistency and confidence in predictions
# Sample learner queries with levels
sample_queries = [
("I want to learn Python from scratch.", "Beginner"),
("I know basic Java and want to improve.", "Intermediate"),
("I want to master deep learning algorithms.", "Advanced"),
("I have some experience with data structures.", "Intermediate"),
("I am new to programming.", "Beginner")
]
print("Sample Learner Queries:\n")
for query, level in sample_queries:
print(f"Query: {query}")
print(f"Level: {level}\n")
Sample Learner Queries:
Query: I want to learn Python from scratch.
Level: Beginner
Query: I know basic Java and want to improve.
Level: Intermediate
Variables
Terminal
2:13 PM
Python 3
29°C
Sunny
Search
ENG
IN
14:23
19-01-2026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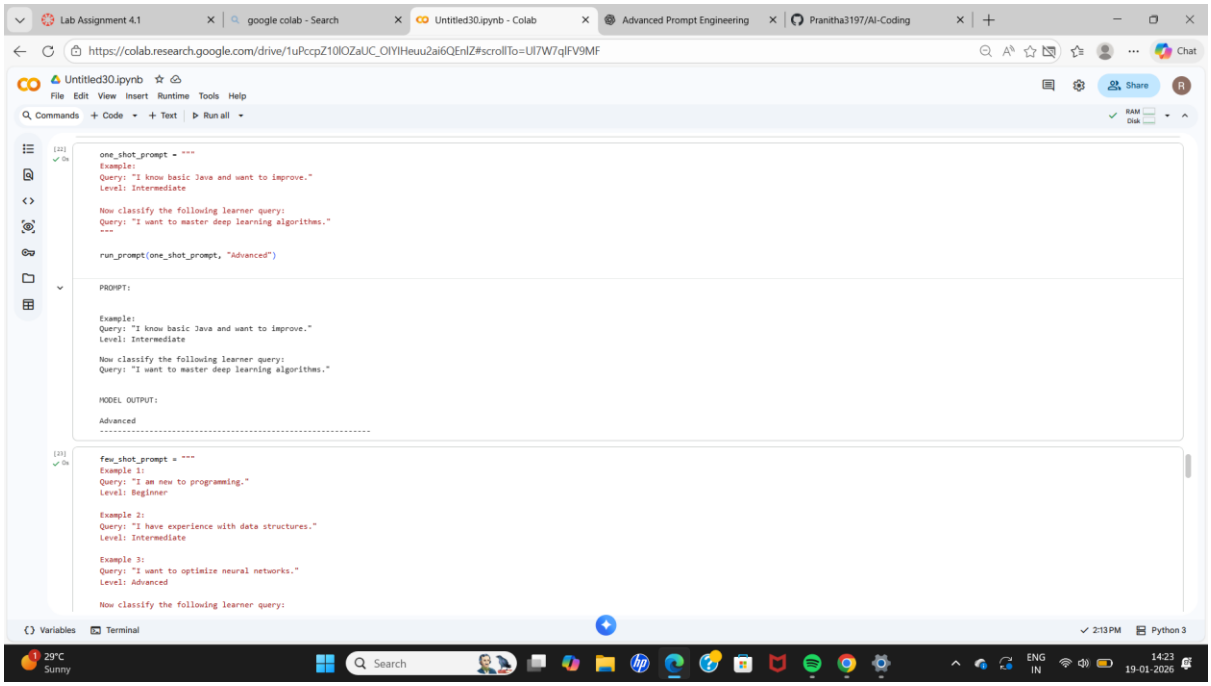
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Query: I want to master deep learning algorithms.
Level: Advanced
Query: I have some experience with data structures.
Level: Intermediate
Query: I am new to programming.
Level: Beginner
def run_prompt(prompt, output):
print("PROMPT:\n")
print(prompt)
print("\nMODEL OUTPUT:\n")
print(output)
print("-" * 60)
zero_shot_prompt = """
Classify the following learner query into one of these levels:
Beginner, Intermediate, or Advanced.
Query: "I want to learn Python from scratch."
"""
run_prompt(zero_shot_prompt, "Beginner")
PROMPT:
Classify the following learner query into one of these levels:
Beginner, Intermediate, or Advanced.
Query: "I want to learn Python from scratch."
MODEL OUTPUT:
Beginner
Variables Terminal
2:13 PM Python 3
29°C Sunny
Search
ENG IN
14:23 19-01-2026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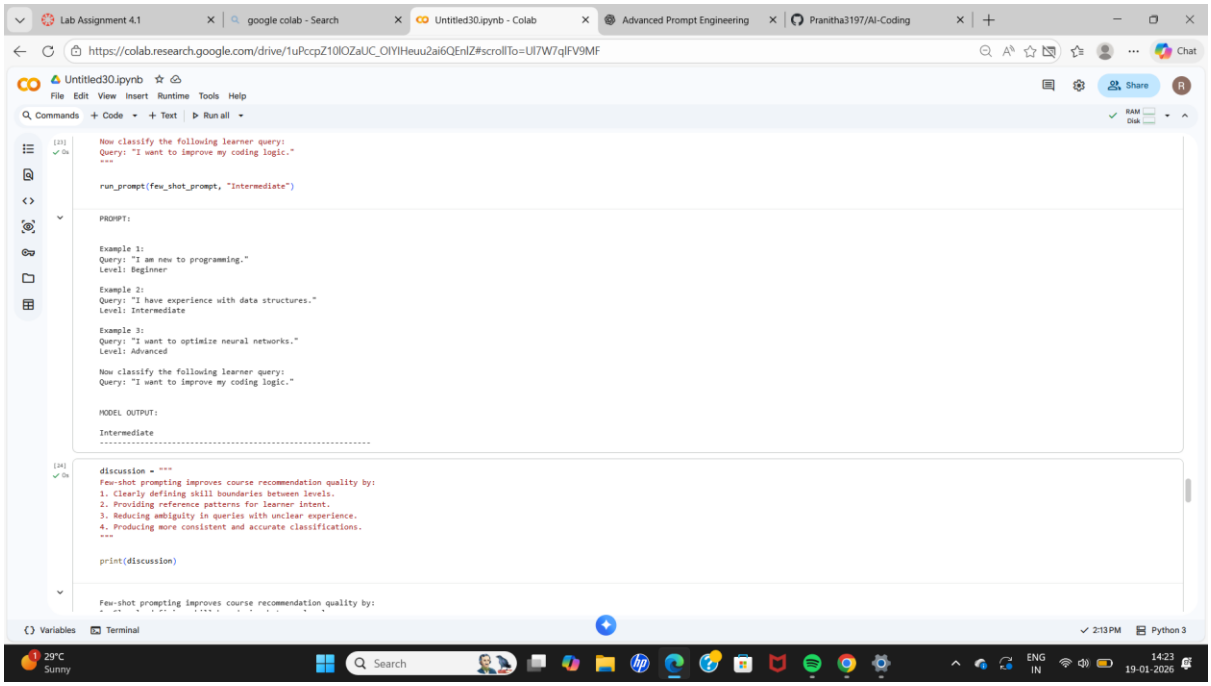
https://colab.research.google.com/drive/1uPccpZ10lOZaUC_OIYIHeuu2ai6QEnlZ#scrollTo=Ul7W7qIFV9MF
Untitled30.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
one_shot_prompt = """
Example:
Query: "I know basic Java and want to improve."
Level: Intermediate
Now classify the following learner query:
Query: "I want to master deep learning algorithms."
"""
run_prompt(one_shot_prompt, "Advanced")
PROMPT:
Example:
Query: "I know basic Java and want to improve."
Level: Intermediate
Now classify the following learner query:
Query: "I want to master deep learning algorithms."
MODEL OUTPUT:
Advanced
few_shot_prompt = """
Example 1:
Query: "I am new to programming."
Level: Beginner
Example 2:
Query: "I have experience with data structures."
Level: Intermediate
Example 3:
Query: "I want to optimize neural networks."
Level: Advanced
Now classify the following learner query:
Variables Terminal
2:13 PM Python 3
29°C Sunny
Search
ENG IN
14:23 19-01-2026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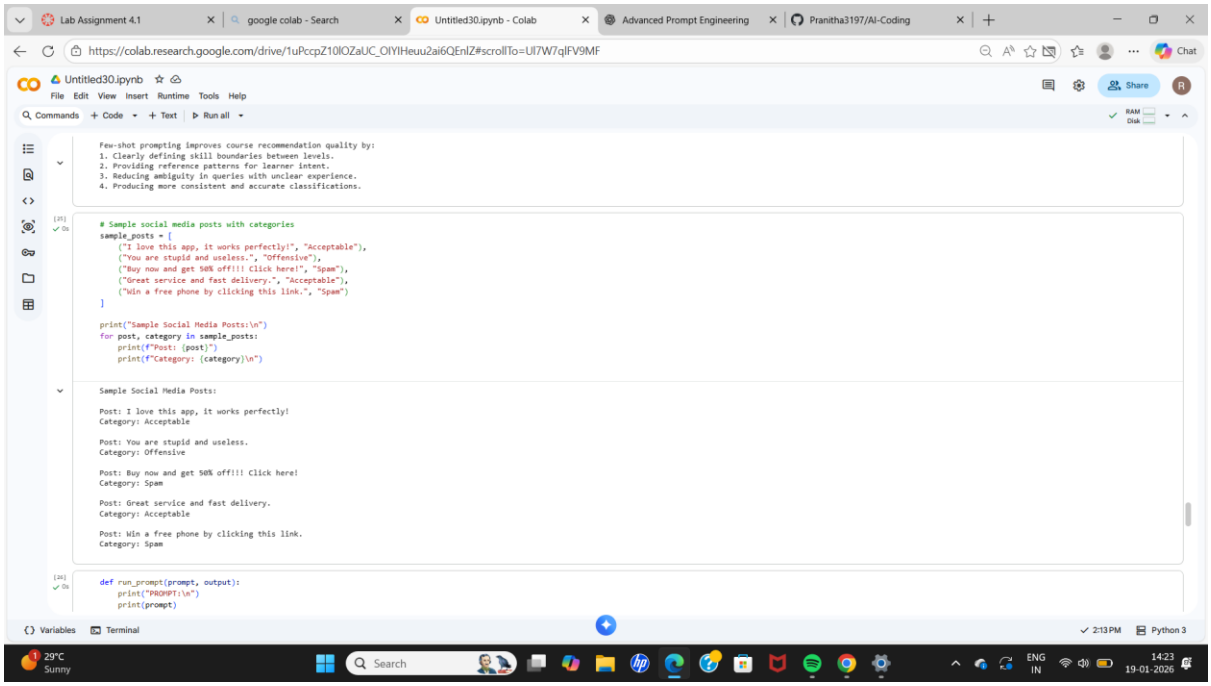
Untitled30.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Share
Now classify the following learner query:
Query: "I want to improve my coding logic."
"""
run_prompt(few_shot_prompt, "Intermediate")
PROMPT:
Example 1:
Query: "I am new to programming."
Level: Beginner
Example 2:
Query: "I have experience with data structures."
Level: Intermediate
Example 3:
Query: "I want to optimize neural networks."
Level: Advanced
Now classify the following learner query:
Query: "I want to improve my coding logic."
MODEL OUTPUT:
Intermediate
discussion = """
Few-shot prompting improves course recommendation quality by:
1. Clearly defining skill boundaries between levels.
2. Providing reference patterns for learner intent.
3. Reducing ambiguity in queries with unclear experience.
4. Producing more consistent and accurate classifications.
"""
print(discussion)
Few-shot prompting improves course recommendation quality by:
Variables Terminal
2:13 PM Python 3
29°C Sunny
Search
ENG IN
14:23 19-01-2026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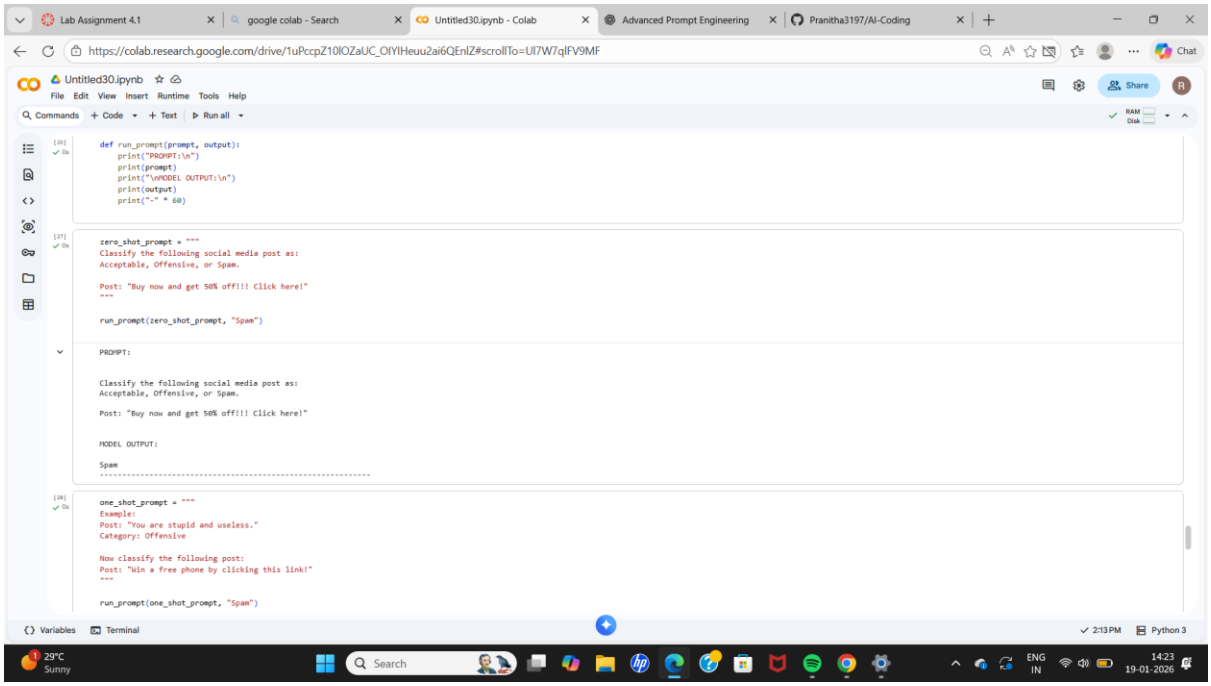
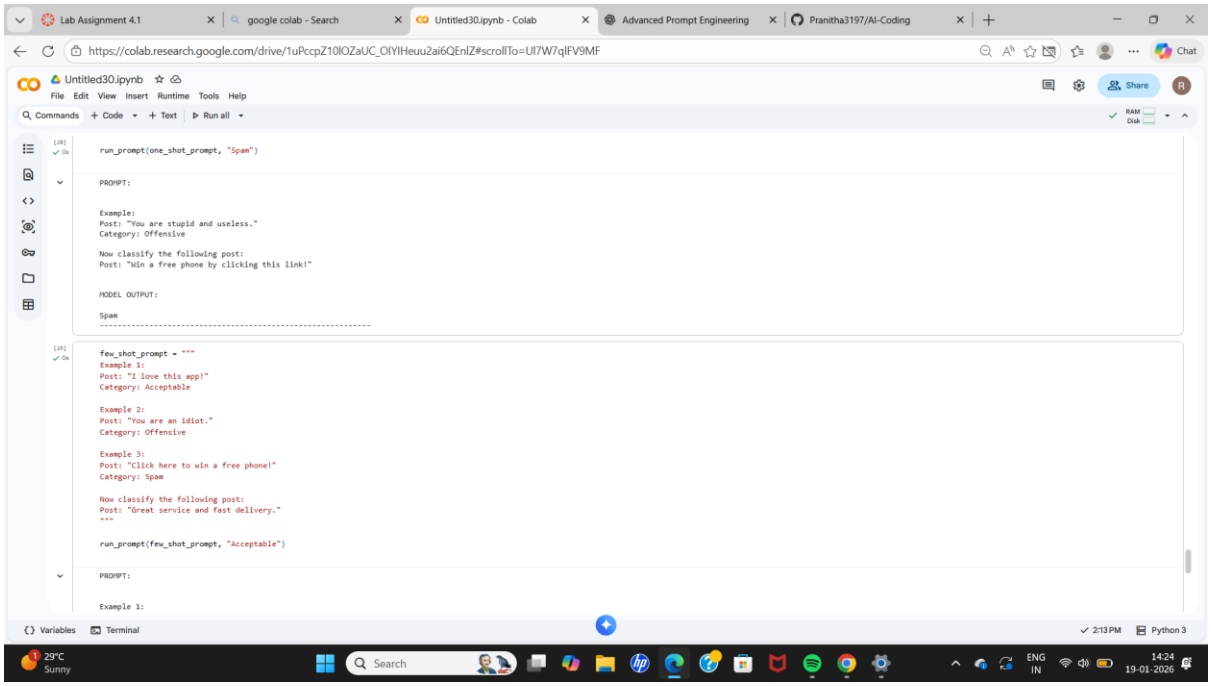
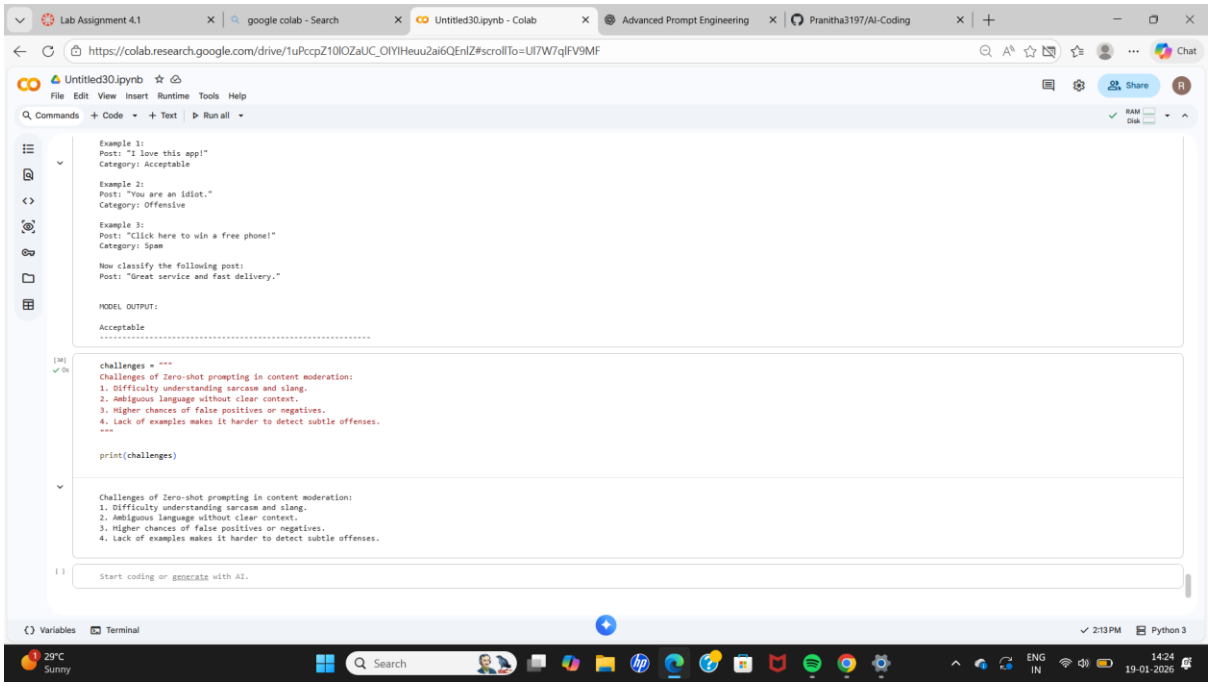
Untitled30.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Share
Few-shot prompting improves course recommendation quality by:
1. Clearly defining skill boundaries between levels.
2. Providing reference patterns for learner intent.
3. Reducing ambiguity in queries with unclear experience.
4. Producing more consistent and accurate classifications.
# Sample social media posts with categories
sample_posts = [
("I love this app, it works perfectly!", "Acceptable"),
("You are stupid and useless.", "Offensive"),
("Buy now and get 50% off!!! Click here!", "Spam"),
("Great service and fast delivery.", "Acceptable"),
("Win a free phone by clicking this link.", "Spam")
]
print("Sample Social Media Posts:\n")
for post, category in sample_posts:
print(f"Post: {post}")
print(f"Category: {category}\n")
Sample Social Media Posts:
Post: I love this app, it works perfectly!
Category: Acceptable
Post: You are stupid and useless.
Category: Offensive
Post: Buy now and get 50% off!!! Click here!
Category: Spam
Post: Great service and fast delivery.
Category: Acceptable
Post: Win a free phone by clicking this link.
Category: Spam
def run_prompt(prompt, output):
print("PROMPT:\n")
print(prompt)
Variables Terminal
2:13 PM Python 3
29°C Sunny
Search
ENG IN
14:23 19-01-2026


Lab Assignment 4.1
google colab - Search
Untitled30.ipynb - Colab
Advanced Prompt Engineering
Pranitha3197/AI-Coding
https://colab.research.google.com/drive/1uPccpZ10lOZaUC_OIYIHeuu2ai6QEnlZ#scrollTo=Ul7W7qIFV9MF
Untitled30.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
def run_prompt(prompt, output):
print("PROMPT:\n")
print(prompt)
print("\nMODEL OUTPUT:\n")
print(output)
print("-" * 60)
zero_shot_prompt = """
Classify the following social media post as:
Acceptable, Offensive, or Spam.
Post: "Buy now and get 50% off!!! Click here!"
"""
run_prompt(zero_shot_prompt, "Spam")
PROMPT:
Classify the following social media post as:
Acceptable, Offensive, or Spam.
Post: "Buy now and get 50% off!!! Click here!"
MODEL OUTPUT:
Spam
one_shot_prompt = """
Example:
Post: "You are stupid and useless."
Category: Offensive
Now classify the following post:
Post: "Win a free phone by clicking this link!"
"""
run_prompt(one_shot_prompt, "Spam")
Variables Terminal
2:13 PM Python 3
29°C Sunny
Search
ENG IN
14:23 19-01-2026


Lab Assignment 4.1
google colab - Search
Untitled30.ipynb - Colab
Advanced Prompt Engineering
Pranitha3197/AI-Coding
https://colab.research.google.com/drive/1uPccpZ10lOZaUC_OIYIHeuu2ai6QEnlZ#scrollTo=Ul7W7qIFV9MF
Untitled30.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
run_prompt(one_shot_prompt, "Spam")
PROMPT:
Example:
Post: "You are stupid and useless."
Category: Offensive
Now classify the following post:
Post: "Win a free phone by clicking this link!"
MODEL OUTPUT:
Spam
few_shot_prompt = """
Example 1:
Post: "I love this app!"
Category: Acceptable
Example 2:
Post: "You are an idiot."
Category: Offensive
Example 3:
Post: "Click here to win a free phone!"
Category: Spam
Now classify the following post:
Post: "Great service and fast delivery."
"""
run_prompt(few_shot_prompt, "Acceptable")
PROMPT:
Example 1:
Variables Terminal
2:13 PM Python 3
29°C Sunny
Search
ENG IN
14:24 19-01-2026



Final Observation for Problem Statement 1

Zero-shot prompting works well for straightforward emails.

One-shot prompting improves understanding by providing context.

Few-shot prompting gives the best performance by clearly defining category boundaries and reducing ambiguity.

Final Observation for Problem Statement 2

Zero-shot prompting works for simple and explicit queries.

One-shot prompting improves understanding with minimal context.

Few-shot prompting provides the best performance by clearly defining intent boundaries and reducing ambiguity.

Final Observation for Problem Statement 3

Zero-shot prompting works for clearly emotional feedback.

One-shot prompting improves understanding with minimal guidance.

Few-shot prompting gives the best accuracy by clearly defining positive, negative, and neutral sentiment patterns

Final Observation for Problem Statement 4

Zero-shot prompting works for very clear beginner or advanced queries.

One-shot prompting improves classification with minimal guidance.

Few-shot prompting provides the best results by clearly distinguishing

between beginner, intermediate, and advanced learning needs.

Final Observation for Problem Statement 5

Zero-shot prompting works for clearly spam or offensive posts.

However, it struggles with ambiguity and sarcasm.

One-shot improves clarity, while Few-shot prompting gives the

most accurate and reliable moderation results.